

UNIT 4: ALGEBRA

Expressions, Equations and Inequalities

Investigation: Work in groups.

Find out as much as you can about the early history of algebra. You might consider:

- a) Ahmes Papyrus (Egyptian c. 1700 BC)
- b) Diophantus (Greek c. AD 250)
- c) Mohammed ibn Musa al-Khowarizmi (Arab c. AD 825)
- d) Bhaskara (Hindu c. AD 1150)

Present your findings to the whole class.

Simplification of expressions

In algebra we are doing arithmetic with just one new feature, we use letters to represent numbers. Because letters are simply stand-ins for numbers, arithmetic is carried out exactly as it is with numbers.

Algebraic expressions are general arithmetic statements where letters are used to represent numbers. For instance

' $x + 5$ ' means 'add 5 to a variable x ' or '5 more than x '

' $a + b$ ' means 'the sum of a and b '

We can also write a statement in form of symbols, for example,

'the product of a and b ' can be written as ' ab '

c is y subtracted from x is $c = x - y$

Exercise 1: In groups,

1. Write in words, the meaning of:
 - a) $c - b$
 - b) $2a + d$

- c) $3(l + m)$
 - d) $p^2 + q$
 - e) $a + b$
2. Write the following as an algebraic expression:
- a) The sum of p and the square of q
 - b) three times the product of a and b
 - c) twice n minus m
 - d) The square of the sum of c and d
 - e) 10 less than c
3. Write in words the meaning of:
- a) $A = 2x + 2y$
 - b) $X = \frac{2c}{d}$
 - c) $B = 3a^2 + d$
 - d) $f = 4x - y$

Simplifying algebraic expressions

Before you evaluate an algebraic expression, you need to simplify it. This will make all your calculations much easier. Here are the basic steps to follow to simplify an algebraic expression:

1. remove parentheses by multiplying factors
2. use exponent rules to remove parentheses in terms with exponents
3. combine like terms by adding coefficients
4. combine the constants

Let's work through an example.

$$5(2 + x) + 3(5x + 4) - (x^2)^2$$

When simplifying an expression, the first thing to look for is whether you can clear any parentheses. Often, you can use the distributive property to clear parentheses, by multiplying the factors times the terms inside the parentheses. In this expression, we can use the distributive property to get rid of the first two sets of parentheses.

$$= 10 + 5x + 15x + 12 - (x^2)^2$$

When a term with an exponent is raised to a power, we multiply the exponents, so $(x^2)^2$ becomes x^4 .

$$= 10 + 5x + 15x + 12 - x^4$$

The next step in simplifying is to look for like terms and combine them. The terms $5x$ and $15x$ are like terms, because they have the same variable raised to the same power -- namely, the first power, since the exponent is understood to be 1. We can combine these two terms to get $20x$.

$$= 10 + 20x + 12 - x^4$$

Finally, we look for any constants that we can combine. Here, we have the constants 10 and 12. We can combine them to get 22.

$$= 22 + 20x - x^4$$

Now our expression is simplified. Just one more thing -- usually we write an algebraic expression in a certain order. We start with the terms that have the largest exponents and work our way down to the constants. Using the commutative property of addition, we can rearrange the terms and put this expression in correct order, like this.

$$= -x^4 + 20x + 22$$

Exercise 2: Simplifying Algebraic expressions

In groups, write the following statements as algebraic expressions and simplify where necessary.

Simplify each of the following by collecting like terms:

(a) $4a + b + 2a$

(b) $4b + 2c + 6b + 3c$

(c) $4a + 5b - a + 2b$

(d) $14p + 11q - 8p + 3q$

(e) $6x - 4y + 8x + 9y$

(f) $11x + 8y + 3z - 2y + 4z$

(g) $16x - 8y - 3x - 4y$

(h) $11y + 12z - 10y + 4z + 2y$

Algebraic Substitution

In Algebra "Substitution" means putting numbers where the letters are:

	When we have:	$x - 2$
	And we know that $x=6$...	
	... then we can "substitute" 6 for x:	$6 - 2 = 4$

Example 1: When $x=2$, what is $\frac{10}{x} + 4$?

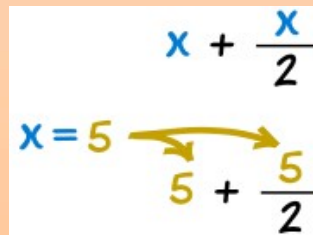
Put "2" where "x" is:

$$\frac{10}{2} + 4 = 5 + 4 = 9$$

Example 2: When $x=5$, what is $x + \frac{x}{2}$?

Put "5" where "x" is:

$$5 + \frac{5}{2} = 5 + 2.5 = 7.5$$



$$x + \frac{x}{2}$$

$$x = 5 \rightarrow 5 + \frac{5}{2}$$

Example 3: If $x=3$ and $y=4$, then what is $x^2 + xy$?

Put "3" where "x" is, and "4" where "y" is:

$$3^2 + 3 \times 4 = 3 \times 3 + 12 = 21$$

Example 4: If $x=3$ (but we don't know "y"), then what is $x^2 + xy$?

Put "3" where "x" is:

$$3^2 + 3y = 9 + 3y$$

(that is as far as we can get)

As that last example showed, we may not always get a number for an answer, sometimes just a simpler formula.

Negative Numbers

When substituting negative numbers, put () around them so you get the calculations right.

Example 1: If $x = -2$, then what is $1 - x + x^2$?

Put "(-2)" where "x" is:

$$1 - (-2) + (-2)^2 = 1 + 2 + 4 = 7$$

In that last example:

- the $-(-2)$ became $+2$
- the $(-2)^2$ became $+4$

Exercise 3

Working in pairs, answer the following questions:

1. Given that $p = 2z - m$, find
 - i) p when $z = 10$ and $m = 2$
 - ii) p when $z = -5$ and $m = 3$
 - iii) m when $p = 20$ and $z = 2$
2. If $p = 4$, $q = -2$ and $r = 3$, find the value of

- i) $3q - 2r$
- ii) $2pq - r$
- iii) $\frac{p - 2q + 2r}{p + r}$

3. If $a = 3, b = -2$ and $c = -1$, find the value of

- i) b^2
- ii) $ab - c^3$

4. If $p = 4, q = -3$ and $r = 2$, find the value of

- i) $\sqrt{p - q + r}$
- ii) $\sqrt{p^2 + q^2}$

5. A formula states:

$$y = 4x - 5$$

- a) Calculate y if $x = 3$.
- b) Determine x if $y = 23$.
- c) Determine x if $y = 8$.

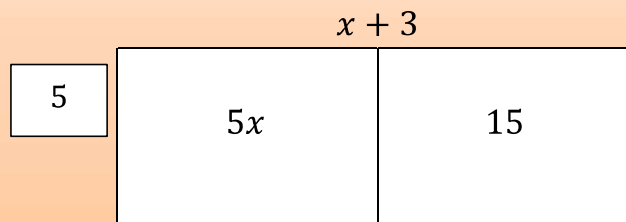
Expansion of algebraic expressions

Expanding is to remove brackets from an algebraic expression.

One bracket

All terms inside the bracket are multiplied by the term outside.

$$5(x + 3) = 5x + 15$$



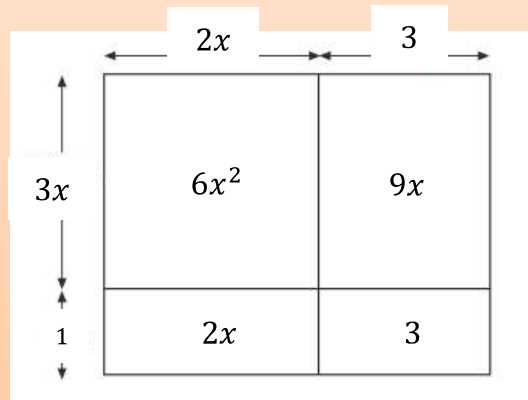
Note that the negative sign belongs to the term directly following it.

e.g. $-3(4a - 5) = (-3) \times 4a + (-3) \times (-5) = -12a + 15$

Multiplying out two brackets

If there are two brackets, multiply everything in the first bracket by everything in the second bracket. To simplify, add the like terms together.

$$(2x + 3)(3x + 1) = 6x^2 + 9x + 2x + 3 = 6x^2 + 11x + 3$$



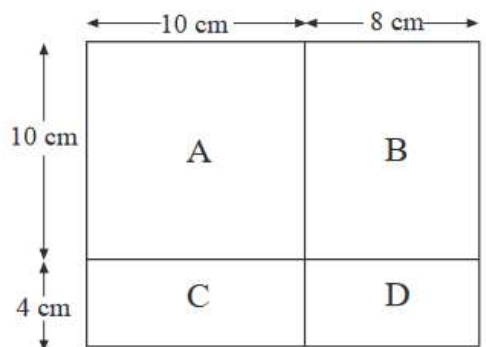
Brackets and squares

If a bracket is squared, write it as bracket multiplied by bracket and expand.

$$\begin{aligned}\text{e.g. } (2x + 3)^2 &= (2x + 3)(2x + 3) \\ &= 4x^2 + 6x + 6x + 9 \\ &= 4x^2 + 12x + 9\end{aligned}$$

Exercise 4: In pairs, work out the following questions.

1. a) The diagram shows a rectangle 18 cm long and 14 cm wide. It has been split into four smaller rectangles, A, B, C and D.
 - i) Write down the area of each of the small rectangles. One has been done for you.



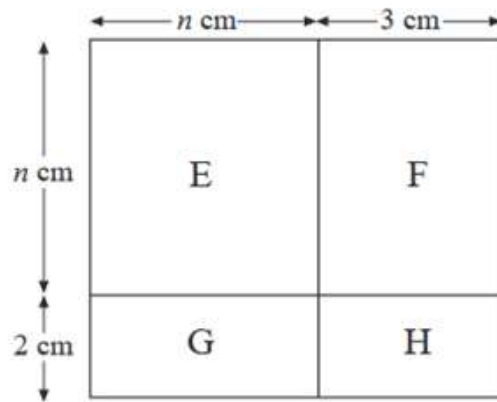
Area of Rectangle C = 40 cm^2 .

ii) What is the area of the *whole* rectangle?

iii) What is 18×14 ?

a) The diagram shows a rectangle $(n + 3)$ cm long and $(n + 2)$ cm wide. It has been split into four smaller rectangles.

a) Write down a number or an expression for the area of each small rectangle. One has been done for you.



Area of Rectangle F = $3n \text{ cm}^2$.

iii) What is $(n + 3)(n + 2)$ multiplied out?

2. Expand:

a. $(x + 6)$

b. $3(x - 4)$

c. $5(2x + 6)$

d. $7(3x - 4)$

e. $3(2x + 4)$

f. $8(3x - 9)$

g. $(-2)(x - 4)$

h. $(-3)(8 - 2x)$

i. $5(3x - 4)$

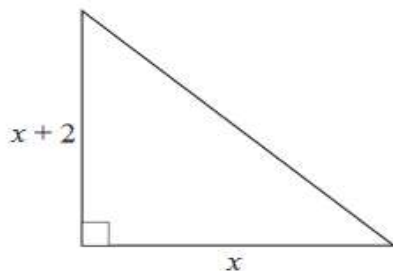
j. $9(2x + 8)$

3. Deng writes $3(4x - 8) = 12x - 8$. Explain why his expansion is *not* correct.

4. Write down an expression for the area of this triangle, that:

a) contains brackets,

b) does *not* contain brackets.



5. Expand each of the following expressions by multiplying out the brackets.

a) $(x + 4)(x + 3)$ c) $(x + 2)(x + 4)$ e) $(x + 1)(x + 5)$

b) $(x + 6)(x - 1)$ d) $(x - 4)(x + 2)$ f) $(x - 3)(x + 2)$

6. Simplify each of the following expressions.

a) $(x + 2)(x + 4) + (x + 1)(x + 2)$

b) $(x + 3)(x + 7) + (x - 1)(x + 5)$

c) $(x + 6)(x + 2) - (x - 2)(x + 3)$

d) $(x - 4)(x - 8) - (x - 1)(x - 9)$

7. Expand.

a) $(x + 1)^2$

i) $(x + 4)^2$

b) $(x - 2)^2$

j) $(2x + 3)^2$

c) $(x + 3)^2$

k) $(4x - 7)^2$

d) $(x + 5)^2$

l) $(3x + 2)^2$

e) $(x - 7)^2$

m) $(4x + 1)^2$

f) $(x - 8)^2$

n) $(5x - 2)^2$

g) $(x + 10)^2$

o) $(6x - 4)^2$

h) $(x - 12)^2$

8. Expand:

b) $(x + 1)^3$

b) $(2x + 1)^3$

c) $(x - 5)^3$

Factorizing algebraic expressions

In this section, we consider examples of the process of factorizing, whereby the process of removing the brackets is reversed and introduced into expressions.

Example 1

Factorise:

a) $8x + 12$

b) $35x + 28$

Solution

a) Note that both terms are multiples of 4, so we can write,

$$8x + 12 = 4(2x + 3)$$

b) Here both terms are multiples of 7, so

$$35x + 28 = 7(5x + 4)$$

Results like these can be checked by multiplying out the bracket to get back to the original expression.

Example 2

Factorise,

a) $x^2 + 2x$

b) $3x^2 - 9x$

c) $x^3 - x^2$

Solution

a) Here, as both terms are multiples of x , we can write,

$$x^2 + 2x = x(x + 2)$$

b) In this case, both terms are multiples of x and 3, giving,

$$3x^2 - 9x = 3x(x - 3)$$

c) In this example, both terms are multiples of x^2 .

$$x^3 - x^2 = x^2(x - 1)$$

Example 3

Factorise:

a) $x^2 + 9x + 18$

b) $x^2 + 2x - 15$

c) $x^2 - 7x + 12$

Solution

- a) This expression will need to be factorized into two brackets.

$$x^2 + 9x + 18 = (x \quad)(x \quad)$$

As the expression begins x^2 , both brackets must begin with x . The two numbers to go into the brackets must multiply together to give 18 and add to give 9. So they must be 3 and 6, giving,

$$x^2 + 9x + 18 = (x + 3)(x + 6)$$

You can check this result by multiplying out the brackets.

- b) We note first that two brackets are needed and that both must contain an x as shown

$$x^2 + 2x - 15 = (x \quad)(x \quad)$$

Two other numbers are needed which, when multiplied give -15 and when added give 2. In this case, these are -3 and 5. So the factorization is,

$$x^2 + 2x - 15 = (x - 3)(x + 5)$$

Check the result by multiplying out the bracket.

- c) Again, we begin by noting that,

$$x^2 - 7x + 12 = (x \quad)(x \quad)$$

We require two numbers which, when multiplied give 12 and when added give -7. In this case, these numbers are -3 and -4.

$$x^2 - 7x + 12 = (x - 3)(x - 4)$$

Exercise 5

1. Factorise:

a) $4x - 2$

b) $6x - 12$

c) $5x - 20$

d) $4x + 32$

e) $6x - 8$

f) $8 - 12x$

g) $21x - 14$

h) $15x + 20$

i) $30 - 10x$

2. Factorise:

a) $x^2 + 4x$

b) $x^2 - 3x$

- c) $4x - x^2$
- d) $6x^2 + 8x$
- e) $9x^2 + 15x$
- f) $7x^2 - 21x$

- g) $28x - 35x^2$
- h) $6x^2 - 14$
- i) $5x^2 - 3x$

3. Factorise:

- i) $x^3 + x^2$
- ii) $2x^2 - x^3$
- iii) $4x^3 - 2x^2$
- iv) $8x^3 + 4x^2$
- v) $16x^2 - 36^3$

- vi) $4x^3 + 22x^2$
- vii) $16x^2 - 6x^3$
- viii) $14x^3 + 21x^2$
- ix) $28x^3 - 49x^2$

- 4. a) Expand $(x + 5)(x - 5)$
- b) Factorise $x^2 - 25$
- c) Factorise each of the following.

- i. $x^2 - 49$
- ii. $x^2 - 64$
- iii. $x^2 - 100$

- ii. $x^2 - a^2$
- iii. $x^2 - 4b^2$

5. Factorise:

- a) $x^2 + 7x + 12$
- b) $x^2 + 8x + 7$
- c) $x^2 + 11x + 18$

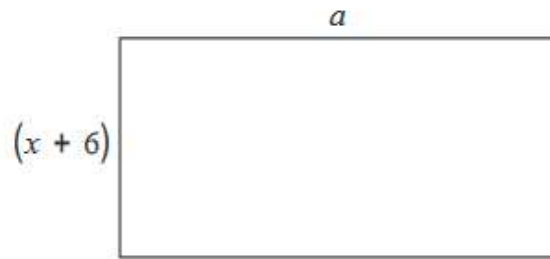
- d) $x^2 + 12x + 27$
- e) $x^2 + 17x + 70$
- f) $x^2 + 6x + 8$

Extension questions

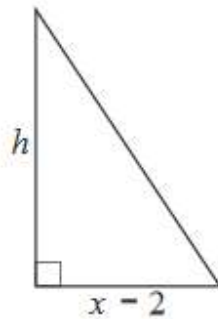
- 6. The area of the rectangle shown is $x^2 - 5x$. Express a in terms of x .



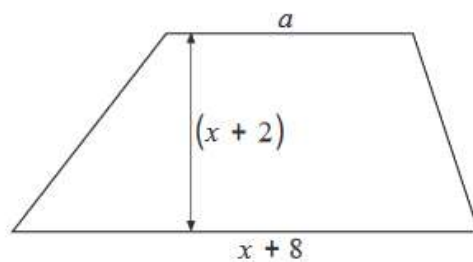
7. The area of the rectangle shown is $x^2 + 11x + 30$. Express a in terms of x .



8. The area of the triangle shown is $\frac{1}{2}x^2 + \frac{3}{2}x - 5$. Express h in terms of x .



9. The area of the trapezium shown is $\frac{1}{2}x^2 + 10x + 18$. Determine a .



Solving Equations

Algebraic equations are equations which contain at least one unknown or variable.

$4x - 7 = x + 5$ is an algebraic equation.

It has a left hand side (LHS) and a right hand side (RHS) separated by an equal sign.

$$\underbrace{4x - 7}_{\text{LHS}} = \underbrace{x + 5}_{\text{RHS}}$$

The solution of an equation is the value or values of the variable which make the equation true.

When the unknown is on both sides, we need to rearrange the equation so that the unknown is on one side and the numbers are on the other side of the equals sign. We do this by doing the same thing to both sides of the equation to keep the equals sign true.

For the equation $4x - 7 = x + 5$, subtract x from both sides

$$\begin{aligned} 4x - 7 - x &= x + 5 - x \\ 3x - 7 &= 5 \end{aligned}$$

Add 7 to both sides

$$\begin{aligned} 3x - 7 + 7 &= 5 + 7 \\ 3x &= 12 \end{aligned}$$

Divide both sides by 3

$$x = 4$$

To check our solution when $x = 4$,

the LHS = $4 \times 4 - 7 = 16 - 7 = 9$ and the RHS = $(4) + 5 = 9$

So, the solution of the equation is $x = 4$.

Simultaneous Equations

Nyoka buys a coffee and cake for SSP. 50.

Othow buys two coffees and a cake for SSP 85.

How much does the coffee cost?

You can figure out that the coffee must cost SSP 35. We can use mathematics to solve problems like these.

So, let the cost of the coffee be x , and the cost of the cake be y .

For Nyoka, $x + y = 50$

For Othow $2x + y = 85$

These equations can be subtracted

$$\begin{aligned} 2x + y - (x + y) &= 85 - 50 \\ x &= 35 \end{aligned}$$

We can find y by substituting for x in the equation for Nyoka or Othow.

$$\begin{aligned} 35 + y &= 50 \\ y &= 50 - 35 = 15 \end{aligned}$$

The solution to these equations is $x = 35$, $y = 15$.

We can check our solution by substituting in the equation for Othow.

$$2x + y = 2 \times 35 + 15 = 85$$

Example

Solve the simultaneous equations

$$4x + 5y = 17$$

$$3x + y = 10$$

Solution

The equation $3x + y = 10$ can be rearranged to give, $y = 10 - 3x$.

This can be substituted into the equation $4x + 5y = 17$ to give

$$4x + 5(10 - 3x) = 17$$

Expanding the brackets, $4x + 50 - 15x = 17$

Simplifying, $50 - 11x = 17$

Add $11x$ to both sides, $50 = 17 + 11x$

Subtract 17 from both sides, $33 = 11x$

Divide both sides by 11, $x = 3$

Substituting for x in equation $y = 10 - 3x$, $y = 10 - 3 \times 3 = 1$

The solution to these equations is $x = 3, y = 1$

We can check by substituting in the original equations, e.g.

$$4x + 5y = 4 \times 3 + 5 \times 1 = 12 + 5 = 17$$

Exercise 6

1. Solve the following equations.

a) $2x - y = 1$
 $2x + 2y = 10$

b) $2x + 3y = 8$
 $5x - 2y = 1$

c) $3x + 3y = 7$
 $4x + 3y = 2$

d) $5x - y = 15$
 $x + y = -3$

e) $3x - y = 6$

$$2x + 5y = -13$$

f) $3x + 5y = 1$
 $x - y = -5$

g) $2x - 5y = 3$
 $3x + 2y = 14$

h) $2x + 3y = 6$
 $3x + 2y = -1$

i) $3x - y = 18$
 $x + 2y = -1$

2. Make up some equations of your own for your friends to try.

Formation and Solution of Inequalities

Equations and inequalities are both mathematical sentences formed by relating two expressions to each other. In an equation the two expressions are deemed equal which is shown by the symbol $=$.

$$x = y \quad x \text{ is equal to } y$$

Where as in an inequality the two expressions are not necessarily equal which is shown by the symbols: $>$, $<$, $\leq$ or $\geq$.

$x > y$	x is greater than y
$x \geq y$	x is greater than or equal to y
$x < y$	x is less than y
$x \leq y$	x is less than or equal to y

An equation or an inequality that contains at least one variable is called an open sentence.

When you substitute the variable in an open sentence with a number the resulting statement is either true or false. If the statement is true the number is a solution to the equation or inequality.

Example 1

Is 3 a solution to

$$5x + 14 = 24$$

Substitute x for 3

$$5 \times 3 + 14$$

$$15 + 14 = 29 \neq 24$$

FALSE!

Since 29 is not equal to 24, 3 is not a solution to the equation.

Example 2

Solve the inequality $2x < 5$.

2 times x is less than 5.

Divide both sides by 2.

x is less than $\frac{5}{2}$.

So the solution is:

$$x < \frac{5}{2}.$$

So x is any number less than $\frac{5}{2}$.

Example 3

Solve the inequality $3x + 1 > 7$.

We want the x term on the left-hand side, by itself.

So, take 1 away from the left-hand side.

To balance the equation, we must do the same to the right hand side.

So, take 1 away from the right-hand side.

This gives us, $3x + 1 - 1 > 7 - 1$

So, $3x > 6$.

$$x > \frac{6}{3} = 2$$

The solution is $x > 2$.

Example 4

Solve the inequality $4a - 2 < 5$.

Add 2 to both sides, to leave $4a$ on the left, by itself:

$$4a - 2 + 2 < 5 + 2$$

So, $4a < 7$.

Therefore, the solution is $a = \frac{7}{4}$.

Example 5

Solve the inequality $2(y + 5) < 16$.

Either:

$$2y + 10 < 16 \text{ (multiplying out the brackets.)}$$

So, $2y + 10 - 10 < 16 - 10$ (subtracting 10 from both sides.)

$$2y = 6$$

Therefore, $y = 3$.

Or:

Divide both sides of $2(y + 5) < 16$ by 2, to get

$$y + 5 < 8.$$

$$\text{Therefore } y = 3$$

Exercise 7

Solve the following inequalities.

1. $4a < 9$
2. $2y + 6 > 14$
3. $3x - 5 < 1$
4. $3(4 + x) < 15$
5. $4x + 20 < 100$
6. $10x + 16 > 156$
7. $25 + 6x < 300$
8. $-5x > 80$